

TABLE II
Some PC-Based Oscilloscopes

Manufacturer	Model	Bandwidth	Sample rate	Multifunction
Dynoninstruments	ELAB-080	60MHz	80MS/s	Yes
Bitscope	BS50	70MHz	40MS/s	Yes
Virtins Technology	VT DSO-2820R	80MHz		Yes
Link Instruments	MSO-19	60MHz	Repetitive mode: 1GS/s Single shot: 200MS/s	Yes
Jyetechn	DSO 094 (Dual channel)	10MHz	50MS/s	No
Parallax	PropScope USB Oscilloscope		25MS/s	Yes
Picotech	PicoScope 2104	10MHz	Repetitive mode: 1GS/s; Single shot: 50MS/s	Yes
Dynoninstruments	S2X100	50MHz	100MS/s	Yes
Easysync	DS1M12	250kHz	Repetitive mode: 20MS/s	Yes
Synchrotech	USB-SCOPE50	75MHz	1GS/s	No

TABLE III
Some Pocket-Size Oscilloscopes

Manufacturer	Model	Bandwidth	Sample rate
Jyetechn	DSO 094 (Dual channel)	10MHz	50MS/s
Velleman	APS230	2 x 30MHz	240MS/s
Velleman	HPS10	2MHz	10MHz
Velleman	HPS50	12MHz	40MHz
Velleman	PPS10	2MHz	10MS/s
uni-trend	UT81B	8MHz	40MS/s
Seedstudio	DSO Nano v2	0-200kHz	1MS/s
Seedstudio	DSO Nano v3	0-200kHz	1MS/s

To capture and debug serial bus you will need record length as high as 1M points.

External trigger. External triggers allow you to stabilise repetitive waveforms and make them appear static on the display by continuously displaying the same section of the input signal. Generally, oscilloscopes are available with edge and pulse-width trigger. But advanced triggering options like A and B sequence, video, communication (CAN, SPI, etc) and logic triggering can help you debug faster.

Waveform capture rate. Sample rate indicates how fast the oscilloscope samples the input signal within one

waveform, but waveform capture rate refers to how quickly an oscilloscope captures a waveform. The oscilloscopes with better waveform capture rate capture fast transients better.

Probe specifications.

Probes' specifications are as important as that of oscilloscope. The probe's bandwidth should match the bandwidth of the oscilloscope. Here also the five time thumb rule works perfectly.

The probe will be in direct contact with the circuit and

should not overload it. Otherwise, the measurements will not be correct. Resistive loading greater than 10M-ohms and capacitive loading less than 10pF is acceptable for a probe.

Integrated display type oscilloscope

Table I show some low-budget integrated-display oscilloscopes available from different manufacturers, including their brief specifications. The bandwidth you can get in this budget will mostly be less than 100MHz, which is more than enough for basic analysis and testing, if you go by 'five times' thumb rule discussed earlier. And you get a lot of

advanced features of expensive oscilloscopes, such as math functions and USB storage, also with these oscilloscopes.

Being mostly compact in size, these can be used as bench-top oscilloscopes and also moved easily, if required. Their display size ranges from 12.7cm to 17.8cm (5 to 7 inches), which is enough for proper viewing. But some vendors boast of utilising the same space more efficiently with advanced features like auto-hide menu panel and sectional zooming.

PC-based oscilloscopes

A PC-based oscilloscope is normally a small piece of hardware that is attached to a computer to acquire the signal and show it on the computer's display. In most cases the power supply too is derived from the computer's USB port, which makes them very handy and easy to carry along. A lot of multifunction options are also available, such as additional signal generator, logic analyser, multimeter, 5V DC supply, digital I/Os and spectrum analyser.

The multifunction oscilloscopes are perfect for students and hobbyists, who do not require higher accuracy but need all the tools at one place. Availability of the signals on a bigger display is also a treat for the eyes. The window software for these oscilloscopes provide analysis and storage tools. Table II shows some of the PC-based systems available in the market.

Pocket-size oscilloscopes

Pocket-size oscilloscopes are easy to carry along and normally work on battery. The frequency range that you get in these oscilloscopes is limited. Some expensive ones do support signals in megahertz range but the low-budget ones are normally for kilohertz range only. So these cannot be used for debugging high-frequency signals; they only support basic testing and are very handy in field work.